

Scaffold-Controlled Evaluation of Molecular Representations for Antimalarial Activity Prediction

Myke Vital Sao Temgoua^{*1}, Jean-Pierre Tchapel Njafa¹, Serge Guy Nana Engo¹, Penabei Samafou², and Wilfred Fon Mbacham³

¹Department of Physics, Faculty of Science, University of Yaoundé I, P.O. Box 812, Yaoundé, Cameroon

²Department of Medical Imaging and Radiation Sciences, Université de Sherbrooke, Sherbrooke, QC, Canada

³Department of Biochemistry, Faculty of Science, University of Yaoundé I, P.O. Box 812, Yaoundé, Cameroon

Abstract

We evaluated whether graph, transformer, and topological molecular representations improve antimalarial activity prediction beyond an ECFP4-random-forest reference under scaffold separation. On 19 836 African antimalarial natural products using five-fold cross-validation with 25 fold-seed replicates, ECFP4-RF achieved ROC AUC 0.830 0, exceeding GIN-TFP (0.813 8), GIN-TNE (0.809 0), GIN (0.804 7), and ChemBERTa (0.786 7); all four comparisons remained significant after Benjamini-Hochberg correction. The transformer evaluation enforces fold-independent pretrained-weight restoration, preventing cross-fold weight transfer from inflating apparent performance. Persistent-image dimensions showed the largest projection-weight salience in the fusion head but did not improve ranking. The same ECFP4-RF-over-GIN ordering held on a molecule-disjoint ChEMBL transfer panel. These results provide a controlled negative benchmark: the conclusions apply to the evaluated data, splits, architectures, and training protocol, not universally to learned molecular models.

Contribution: We provide a controlled comparison of graph, sequence, and topological molecular representations under scaffold separation, using ECFP4-RF as a reference model. The evaluation documents the fold-independent initialization protocol that valid transformer cross-validation requires, and tests topological and tensor-network descriptors inside a graph architecture under scaffold-held-out evaluation on an antimalarial natural-product panel.

Keywords: graph neural networks; molecular fingerprints; antimalarial natural products; scaffold split; benchmark; interpretability; persistent homology

Highlights:

- Under the scaffold split, ECFP4-RF achieved the highest AUC (0.830 0); each learned arm was lower after BH correction (adjusted $p \leq 0.037$ 8)
- Fold-independent transformer initialization is required for valid cross-validation and is made explicit in this benchmark
- Persistent-image dimensions were most prominent in the fusion projection weights, without evidence of predictive improvement
- Five-fold evaluation repeated over five seeds distinguishes split effects from variation across training runs

^{*}Corresponding author: myke-vital.sao@facsciences-uy1.cm

- The same ECFP4–RF-over-GIN ordering holds on a molecule-disjoint external ChEMBL panel

1 Introduction

Malaria remains a leading infectious disease burden in sub-Saharan Africa, with 247 million cases and 619 000 deaths in 2023 [1]. Control is increasingly threatened by the emergence of artemisinin-resistant *Plasmodium falciparum* strains in Southeast Asia and Africa [2, 3], prompting an urgent, WHO-endorsed need for new scaffolds that remain active against drug-resistant parasites. African natural products (NPs) constitute a rich source of chemically diverse scaffolds [4, 5]. The panel studied here descends from a screening campaign seeded on 396 African natural products and 454 synthetic-drug scaffolds [6]. Computational screening of such libraries can accelerate hit identification, yet the choice of molecular representation directly conditions the apparent performance and chemical generalization of the predictor. In recent years, a “scale-centric” narrative has emerged in molecular machine learning: pretrained molecular foundation models and sequence transformers are expected to supersede compact, task-trained cheminformatics models [7, 8]. This narrative extends to quantum-inspired representations, where quantum computing is increasingly discussed as a paradigm for drug discovery and bioinformatics [9, 10].

Recent controlled benchmarks instead indicate that representation scale does not guarantee improved molecular prediction. In a 156-comparison benchmark across 26 endpoints, Guo and Ding found that classical machine learning on fingerprints wins 47.4 % of task–metric entries, ahead of pretrained sequence models (28.8 %) and GNNs (21.8 %), with the classical advantage largest under random splits and shrinking, but not vanishing, under scaffold-separated protocols [11]. An independent comparison of 25 pretrained embedding models across 25 datasets found that *almost no neural embedding statistically improves over the ECFP fingerprint baseline* [12]. And for natural products specifically, a 20-fingerprint benchmark over 100 000+ NPs showed that no single encoding dominates, and that ECFP is frequently matched or exceeded by alternative fingerprints on NP bioactivity tasks [13].

We evaluated whether learned models improve on ECFP4 for this panel, whether persistent-homology and tensor-network descriptors reduce the scaffold-generalization gap when fused with GIN, and how fold-independent initialization affects the evaluation of a pretrained transformer.

1.1 Related work

The tension between classical cheminformatics and neural molecular learning has been documented across multiple benchmarks. For molecular property prediction, random forests on ECFP fingerprints remain competitive with or superior to GNNs on small-to-medium datasets (< 10 000 molecules) [11, 12]. GNN architectures (GCN, GAT, GIN) have shown gains on large, curated datasets such as MoleculeNet [14], but these gains often diminish under scaffold splits that better reflect real-world drug discovery [11]. Pretrained sequence transformers (ChemBERTa, MolBERT) draw on large unlabeled corpora but have shown inconsistent gains on downstream tasks, particularly when the target dataset is structurally dissimilar to the pretraining data [12, 15].

Topological data analysis (TDA) has been applied to molecular representation through persistent homology, producing descriptors that capture ring systems, molecular cavities, and higher-order connectivity [16]. Recent work has fused TDA features with GNN architectures, reporting modest improvements on specific tasks [17]. Multimodal fusion of graph and sequence representations, for example concatenating GCN embeddings with ChemBERTa latent vectors, has shown promise on small datasets [18]. More broadly, computational MoA analyses emphasize that chemical, phenotypic, and network data capture complementary biological levels; target

prediction alone is therefore not equivalent to a complete mechanism assignment [19]. Tensor-network embeddings, derived from Tucker decomposition of molecular tensors, provide compressed representations that retain scaffold-relevant information [8]. However, neither TDA nor tensor-network features have been systematically tested inside GNN fusion architectures under rigorous scaffold-split protocols on antimalarial panels.

We benchmarked GIN, GIN-TFP, GIN-TNE, and ChemBERTa against ECFP4-RF under random and scaffold splits. The analysis separates predictive performance from projection-weight salience and evaluates the effect of fold-independent transformer initialization. We prespecified ROC AUC as the primary ranking metric; AUPRC is reported for the extended fold-level robustness analyses, whereas calibration and nearest-neighbour extrapolation metrics were not part of the locked primary analysis.

2 Results

2.1 ECFP4-RF defines the reference performance

ECFP4-RF provided the reference against which all learned representations were evaluated. Under the random split it achieved 0.9433 ± 0.0003 across five per-seed means (range 0.9428–0.9437), consistent with the previously reported performance for this molecular collection (0.9475 ± 0.0045) despite the distinct frozen split and random-seed realization [20]. This agreement supports comparability of the panel and fingerprint implementation.

2.2 Primary molecular benchmark and orthogonal phenotype analysis

The molecular benchmark was the primary analysis. We also evaluated a phenotype-only baseline on the public LISH mechanism-of-action (MoA) training release as an orthogonal pharmacological reference. This task uses gene-expression, viability, dose, and time features rather than molecular structures; because no versioned drug-SMILES mapping was available, LISH cannot test transfer of ECFP4, GNN, or transformer representations. With vehicle controls retained, the phenotype-only baseline achieved mean column-wise log loss 0.02378 and macro-AUROC 0.6435 across 25 fold-seed evaluations, and removing vehicle controls changed macro-AUROC by only -0.0018 (full protocol, complete metric set, and sensitivity table in the Supporting Information (Section 1)). These metrics are not numerically comparable with the molecular ROC-AUC results because the task, labels, features, aggregation unit, and primary metric differ; the labels indicate MoA-associated prediction rather than causal target engagement [19].

2.3 ECFP4-RF outperforms learned representations under scaffold separation

Table 3 reports the mean test ROC AUC and the SD across five per-seed means for each model and split; each per-seed mean aggregates the five fold-level test AUCs. Under the scaffold split, the fingerprint baseline reaches 0.8300. Every graph-based and transformer arm is significantly *worse*: the best GNN arm, GIN-TFP, reaches 0.8138 ($\Delta = -0.016$, raw paired $p = 0.025$; BH-adjusted $p = 0.0330$); GIN-TNE reaches 0.8090 ($\Delta = -0.021$, raw $p = 0.038$; BH-adjusted $p = 0.0378$); plain GIN reaches 0.8047 ($\Delta = -0.025$, raw $p = 0.018$; BH-adjusted $p = 0.0330$); ChemBERTa reaches 0.7867 ($\Delta = -0.043$, raw $p < 0.0001$; BH-adjusted $p = 0.0002$). All four scaffold comparisons survive Benjamini-Hochberg correction applied within that four-test family. Under the random split the same verdict holds: ChemBERTa reaches 0.9121 ($\Delta = -0.031$, raw $p < 0.0001$), GIN 0.9098 ($\Delta = -0.034$, raw $p < 0.0001$), GIN-TFP 0.9084 ($\Delta = -0.035$, raw $p < 0.0001$), and GIN-TNE 0.8918 ($\Delta = -0.052$, raw $p < 0.0001$), each significantly below ECFP4-RF (0.9433); all four random comparisons survive

Table 1: Summary of the P5 benchmark design.

Element	Specification
Panel	19 836 labelled molecules
Task	Binary antimalarial activity prediction
Reference	ECFP4-RF (2 048-bit fingerprint; 500 trees)
Learned models	GIN, GIN-TFP, GIN-TNE, ChemBERTa
Splits	Stratified random and Bemis-Murcko scaffold
Replication	Five folds $\times$ five seeds per split
Primary metric	Test ROC AUC
Statistical unit	Five per-seed means, paired by seed
Multiplicity	Benjamini-Hochberg within each four-model family
External analysis	Molecule-disjoint ChEMBL transfer panel

Table 2: Scaffold-split effects relative to ECFP4-RF. Values are differences in mean ROC AUC; paired tests use the five per-seed means ($df = 4$).

Model	Mean ROC AUC	Δ vs. ECFP4-RF	BH-adjusted p
ECFP4-RF	0.830 0	—	—
GIN	0.804 7	−0.025	0.033 0
GIN-TFP	0.813 8	−0.016	0.033 0
GIN-TNE	0.809 0	−0.021	0.037 8
ChemBERTa	0.786 7	−0.043	0.000 2

Benjamini-Hochberg correction applied within their own separate four-test family. *No graph-based arm outperforms the fingerprint baseline under either split.* A distribution-free check corroborates these parametric tests: paired permutation of the five per-seed mean AUCs against ECFP4-RF ($B = 10\,000$ permutations) returns a negative mean Δ AUC for every arm on both splits (two-sided $p = 0.009$ throughout), with bootstrap intervals that exclude zero — on the scaffold split, GIN −0.025 3 (−0.036 4—−0.013 8), GIN-TFP −0.016 2 (−0.025 4—−0.009 9), GIN-TNE −0.021 0 (−0.035 1—−0.012 5), ChemBERTa −0.043 3 (−0.047 5—−0.039 2).

This comparison used a fixed training protocol with AdamW, validation-based early stopping, selection of the validation-optimal epoch, and 25 fold-seed replicates. It is consistent with reports across ADMET/Tox21 endpoints [11] and pretrained-embedding benchmarks [12], in which local substructure encodings captured by ECFP4 and exploited by tree ensembles remain highly data-efficient.

2.4 ChemBERTa does not close the scaffold-generalization gap

ChemBERTa, fine-tuned with per-fold weight reset (see Methods), reaches $0.786\,7 \pm 0.005\,4$ under the scaffold split (SD across five per-seed means; five folds repeated over five seeds; fold-level range 0.724–0.836). The wide fold-level range indicates scaffold-dependent transformer performance, with some test scaffolds approaching the fingerprint baseline. Against the 0.830 0 fingerprint baseline this is a significant deficit ($\Delta = -0.043$, paired $t(4) = -18.35$, $p < 0.000\,1$). Under the random split it reaches $0.912\,1 \pm 0.001\,2$, narrowly ahead of GIN (0.909 8) but still 0.031 below ECFP4-RF (0.943 3). The final scaffold hierarchy is thus:

$$\begin{aligned} \text{ECFP4-RF (0.8300)} &> \text{GIN-TFP (0.8138)} > \text{GIN-TNE (0.8090)} \\ &> \text{GIN (0.8047)} > \text{ChemBERTa (0.7867)}. \end{aligned} \tag{1}$$

The $\pm$ values reported throughout are population standard deviations across per-seed mean AUCs under the fixed five-fold protocol; fold-level dispersions are correspondingly larger. These

Table 3: Mean test ROC AUC $\pm$ SD of the five per-seed means (each seed averages five fold-level test AUCs). The underlying analysis contains 25 fold-seed replicates per split. The displayed p -values are raw paired tests on the five per-seed means versus ECFP4-RF under scaffold; Benjamini-Hochberg correction across the four comparisons gives adjusted p -values of 0.033 0 (GIN), 0.033 0 (GIN-TFP), 0.037 8 (GIN-TNE), and 0.000 2 (ChemBERTa). The GIN-TNE comparison is the weakest of the four (adjusted $p = 0.0378$) but survives correction. All statistical comparisons are documented in the Methods.

Model	Random	Scaffold	Δ vs. ECFP4 (scaffold)
ECFP4-RF (baseline)	0.943 3 $\pm$ 0.000 3	0.830 0 $\pm$ 0.002 3	—
GIN	0.909 8 $\pm$ 0.002 2	0.804 7 $\pm$ 0.014 1	-0.025 ($p = 0.018^*$)
GIN-TFP (topological fusion)	0.908 4 $\pm$ 0.001 2	0.813 8 $\pm$ 0.010 7	-0.016 ($p = 0.025^*$)
GIN-TNE (tensor fusion)	0.891 8 $\pm$ 0.001 8	0.809 0 $\pm$ 0.014 9	-0.021 ($p = 0.038^*$)
ChemBERTa	0.912 1 $\pm$ 0.001 2	0.786 7 $\pm$ 0.005 4	-0.043 ($p < 0.000 1^{***}$)

uncertainty summaries quantify repeated computational runs, not uncertainty over a future population of compounds or datasets. Because the five per-seed means are the inferential unit, the resulting intervals should not be read as population-level confidence intervals or as independent evidence from 25 observations. The ordering among learned models is descriptive; the prespecified inferential comparisons were restricted to each learned arm versus ECFP4-RF.

Validation trajectories across 50 training epochs document the optimization path under the prespecified patience of 10 (Figure 2). All five molecular arms (GIN, GIN-TFP, GIN-TNE, ChemBERTa canonical, ChemBERTa extended) are shown under both random and scaffold splits; each curve aggregates five folds repeated over five seeds. Early stopping selects the validation-optimal epoch before test evaluation; these trajectories describe optimization behavior rather than establishing the absence of over-fitting.

2.5 Topological descriptors remain visible without improving ranking

Even when predictive gains are absent, learned fusion weights carry structure that can be examined descriptively. We compute a *salience* score for each descriptor dimension as the mean absolute weight in the fusion projection layer (head_0) over the 25 replicates. For the TFP fusion (78 dimensions: 33 homology-dimension-0, 25 persistent-image, 20 Betti), the *persistent-image block (dims 33–57) has the largest raw mean projection-weight magnitude* (0.079 0) versus homology-0 (0.048 5) and Betti (0.054 7) blocks; the top five dimensions are persistent-image (42, 43, 52, 53, 54). For the TNE fusion (192 dimensions) the top dimensions are 68, 43, 92, 66, 165, and the top-10 % of dimensions account for 17.3 % of total salience (moderate sparsity).

This pattern indicates that topological descriptors are used by the fusion head, but it does not establish feature necessity or causal relevance. On this panel, the descriptor signal does not translate into a ranking advantage over fingerprints. Thus, projection-weight salience and predictive ranking should be treated as complementary, not interchangeable, summaries.

3 Discussion

Across the three questions, learned representations did not exceed ECFP4 under either split. Topological fusion narrowed the scaffold gap modestly but did not close it, while persistent-image dimensions remained prominent in the fusion head without yielding a ranking advantage. Predictive utility and representation-level visibility are therefore separable properties. The separate LISH phenotype-only benchmark provides complementary pharmacological context, but the absence of a structure mapping prevents assessment of the molecular representations

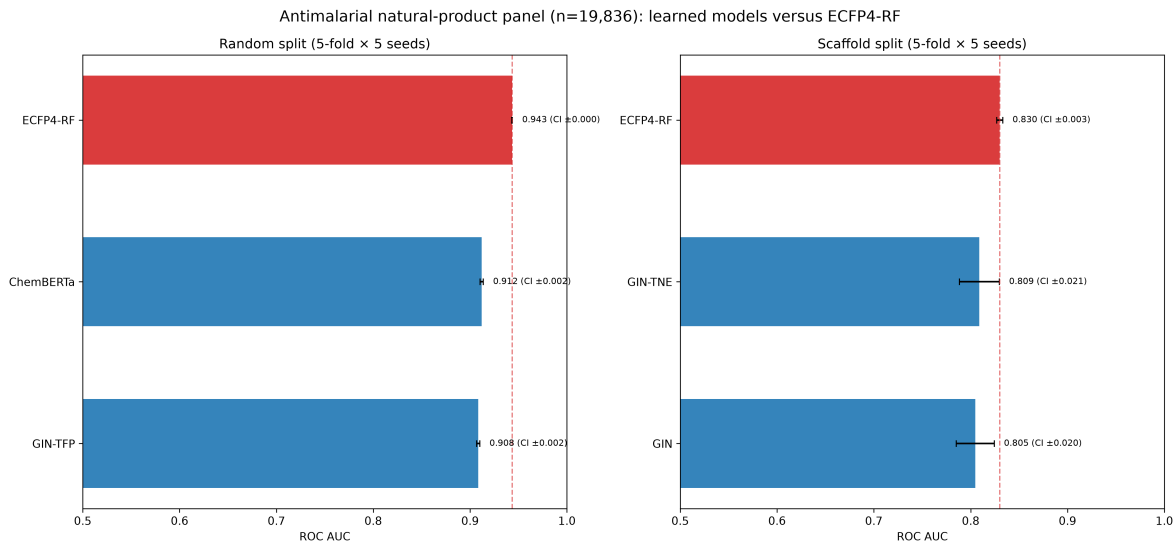


Figure 1: Mean test ROC-AUC with 95 % confidence intervals computed from five per-seed means (each mean aggregates five fold-level test AUCs), shown for random (left) and scaffold (right) splits. The underlying analysis contains 25 fold-seed replicates per model and split. Bar annotations report the mean followed by the CI half-width. The ECFP4-RF baseline (red) is not exceeded by any tested GNN or transformer arm.

studied here; its metrics are not numerically comparable to the molecular ROC-AUC results (Supporting Information (Section 1)).

3.1 Performance depends on representation-task alignment

The tested pretrained and graph-based models did not improve ROC-AUC over ECFP4-RF on this panel, including under scaffold-held-out testing. This conclusion is supported by repeated evaluation, paired comparisons on the five per-seed means, and correction for multiplicity. The transformer result depends critically on fold-independent weight initialization: pretrained weights were reset at the beginning of every fold so that fine-tuning could not transfer between test folds. This initialization detail should be reported explicitly in molecular benchmarks.

3.2 Why fingerprints hold: local substructure dominance

The consistent ECFP4 advantage is compatible with representation-task alignment. ECFP4 encodes local substructure environments, which tree ensembles can exploit efficiently in this labelled natural-product panel [21, 22]. The fusion models show that topological descriptors carry detectable information, but their marginal contribution was insufficient to improve ranking. This interpretation is consistent with broader benchmark evidence that classical fingerprint models often remain competitive with pretrained and graph-based models, while performance depends on the endpoint and split protocol [11, 12]. It is an explanation compatible with the data, not a demonstration that fingerprints are universally optimal. Scaffold splits also remain proxies for novelty and may differ from cluster-based evaluations [23].

3.3 Relation to prior antimalarial modelling

The molecular collection and fingerprint implementation were previously characterized in a quantum-representation benchmark [20]. That study found no statistically detectable advantage of the simulated quantum kernel over a tuned radial-basis kernel; the present analysis tests a different question: whether graph or sequence models improve activity ranking on the same

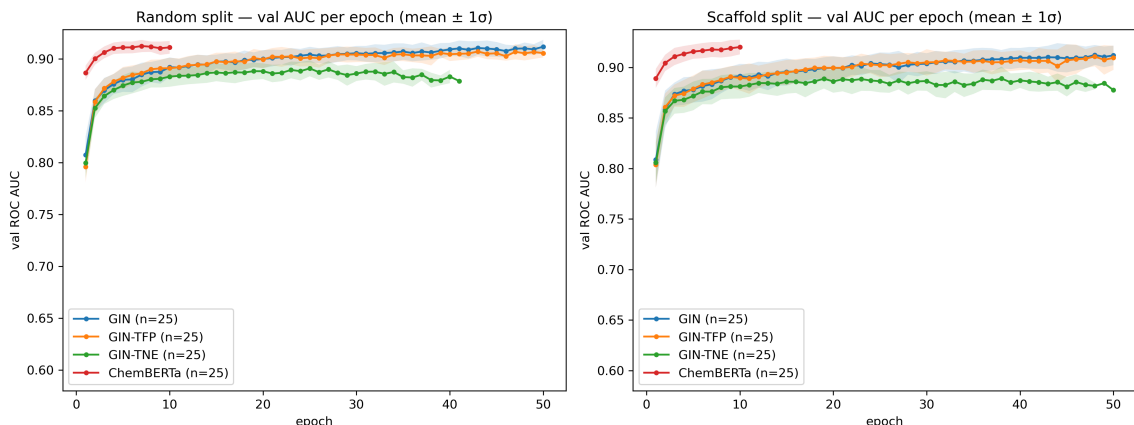


Figure 2: Validation ROC-AUC trajectories used to select the validation-optimal epoch. Curves are available for every tested arm under both random and scaffold splits; each available arm contains five folds repeated over five seeds. Early stopping with patience 10 selects the validation-optimal epoch before test evaluation.

chemical collection. Molecular-dynamics studies of selected antimalarial leads have additionally examined resistance-associated targets and mutations [24]; those simulations provide biological context but are not labels or validation data for the present benchmark. Topological descriptors are used descriptively by the fusion head, but none of the tested learned representations improves ROC-AUC over ECFP4-RF.

3.4 Position relative to recent representation benchmarks

Broad benchmark evidence already shows that compact fingerprint models remain competitive: in a 156-comparison benchmark, classical machine learning on fingerprints won the largest share of task-metric entries [11]; in a 25-embedding comparison, almost no neural embedding statistically improved over ECFP [12]; and in a natural-product-focused comparison, ECFP was frequently matched or exceeded by other fingerprints [13]. The present study does not merely restate those aggregates. Three elements distinguish it. First, the transformer comparison enforces fold-independent restoration of pretrained weights, removing the cross-fold weight-transfer channel that can inflate apparent transformer performance; this initialization protocol is rarely reported in comparable benchmarks. Second, the topological-fusion arms (GIN-TFP, GIN-TNE) test persistent-homology and tensor-network descriptors inside a graph architecture under scaffold-held-out evaluation on an antimalarial natural-product panel, a combination not covered by the aggregate comparisons cited above. Third, the ordering is corroborated on a molecule-disjoint external ChEMBL panel, across three additional scaffold partitions, and with a fully released reproducibility package. The negative result is therefore a controlled result for this panel and protocol, not a restatement of aggregate statistics.

3.5 Implications for molecular-model selection

For comparable antimalarial natural-product panels, ECFP4-RF is a strong reference model: it is computationally inexpensive and outperformed every learned arm tested here. Graph and sequence models should be evaluated against this baseline under scaffold-held-out testing rather than assumed to improve from model scale alone. Transformer studies should reset pretrained weights for every fold and record the initialization protocol. Topological or tensor-network descriptors can be examined as complementary descriptive features, but their inclusion should not be interpreted as evidence of improved ranking without a paired comparison.

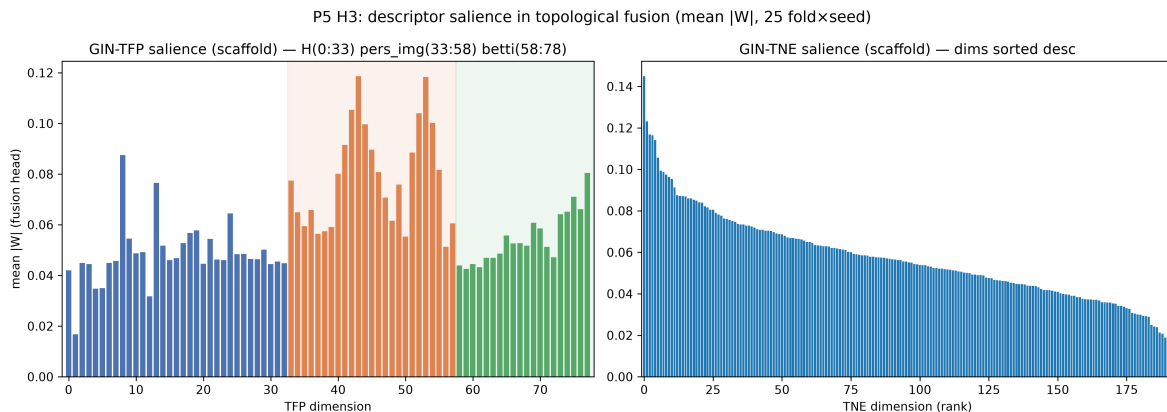


Figure 3: Descriptive projection-weight salience in the topological fusion heads under the scaffold split. Values are mean raw $|W|$ across 25 fold-seed replicates. Left: TFP (78 dimensions), partitioned into homology-0 (dims 0–32), persistent-image (dims 33–57), and Betti (dims 58–77); the persistent-image block has the largest raw weight magnitude. Right: TNE (192 dimensions), shown in descending order. Weight magnitude is interpreted descriptively and is not a causal attribution.

3.6 Robustness analyses

The analyses below test whether the primary ordering depends on a single partition or label source. They extend the evidence base but do not replace the primary frozen-split comparison.

3.6.1 External ChEMBL transfer

On a molecule-disjoint ChEMBL-derived activity benchmark based on recorded IC_{50}/EC_{50} activities against *Plasmodium falciparum* (target ChEMBL364; compounds were labelled active at $pChEMBL \geq 6.0$ and inactive at $pChEMBL < 5.0$, with the intervening 5.0–6.0 band discarded as ambiguous, retaining 22 447 of 31 649 unique compounds, and the 180 molecules sharing a canonical SMILES with the P5 panel were then removed, leaving 22 267 evaluation compounds). The fingerprint baseline again wins: ECFP4-RF 0.9190 ± 0.0004 versus GIN 0.8843 ± 0.0021 under the scaffold split ($\Delta = +0.035$, paired $p < 0.0001$). The fingerprint advantage is even larger on the external panel ($\Delta = 0.035$ versus $\Delta = 0.025$ on the canonical P5 panel), indicating that the ordering holds beyond the eOS80CH screening labels. This is a molecule-disjoint external transfer panel within the same broad antimalarial domain, not a cross-domain or prospective validation. Because this external analysis used an independently constructed split rather than the frozen internal fold files, it is a transfer analysis rather than a strict replication.

3.6.2 GNN capacity sensitivity

The evaluated GNN and fusion models use a single-layer prediction head and fixed hyperparameters; alternative architectures or tuning could reduce the observed gap. A tagged sensitivity run on the frozen scaffold split bracketed the canonical capacity setting (hidden 64/dropout 0.2 and hidden 256/dropout 0.1; Supporting Information (Section 3): scaffold ROC AUC moved only within 0.8000–0.8081 versus the canonical 0.8047, and every configuration remained below the ECFP4-RF reference (0.8300). Accordingly, the negative result applies to the evaluated configurations rather than to all possible GNN or transformer implementations, and it is not an artifact of one specific capacity choice. ChemBERTa was fine-tuned with a deliberately short fixed schedule (10 epochs, patience 3, learning rate 2×10^{-5} , batch size 32) suited to this low-data regime rather than per-task optimization; published fine-tuning practice for chemical language

models reports optimal learning rates near 3×10^{-5} , batch sizes of 16–32, schedules of at least 50 epochs under early stopping, and multi-seed variance reporting [25]. The transformer arm should therefore be read as a compute-matched baseline rather than a maximal-effort implementation, with multi-seed replication left as future work if tighter variance bounds are required. The GIN uses 2D molecular graphs without conformational information; 3D-equivariant architectures may perform differently on tasks for which shape is important [18]. The salience measure (mean absolute projection weight) captures correlation between descriptor dimensions and the predictive head, not causation; more sophisticated methods (e.g., integrated gradients, SHAP) would be needed to establish which features are truly necessary.

3.6.3 Extended scaffold partitions

Finally, the scaffold split is a proxy for novelty. The primary benchmark uses one frozen scaffold partition; a recent structural-frontier analysis shows that splits reserving the sparsest, physicochemically remote scaffold groups inflate error further on ADMET tasks [26]. In a secondary robustness campaign, we generated three additional independent scaffold partitions and retrained the five GNN configurations (GIN, native and permuted GIN-TFP, and native and permuted GIN-TNE) on the canonical panel and on each new partition. Each configuration contained 25 fold-seed records with stored test probabilities, AUPRC, and individual salience vectors. Across the three new partitions, native GIN scaffold ROC AUC ranged from 0.7886 to 0.7967, native GIN-TFP from 0.8013 to 0.8105, and native GIN-TNE from 0.7976 to 0.8008; native AUPRC ranged from 0.9025 to 0.9160. These secondary estimates preserved the negative ordering relative to the canonical ECFP4-RF reference, but they do not replace the prespecified primary table. Descriptor permutation reduced mean AUC numerically for GIN-TFP on all three new partitions (mean differences native minus permuted 0.0054–0.0139) and for GIN-TNE on two of three partitions (0.0053–0.0092); exact sign-flip tests on five per-seed means were not significant after accounting for the small computational unit. The extended campaign is consistent with a modest contribution from the native descriptor arrays, although the available paired tests do not establish that contribution. Salience stability is now computable from the stored vectors: across the canonical scaffold and three new partitions, mean pairwise Spearman correlations were 0.755–0.829 for native TFP and 0.789–0.835 for native TNE, while mean top-10% Jaccard overlap was 0.374–0.399 and 0.274–0.326, respectively. These are stability diagnostics for weight patterns, not causal feature attributions. The standardization audit found no invalid SMILES or exact/parent duplicate rows in the 19836-molecule panel, but identified 306 tautomer-collision rows; the primary labels were not changed. The frozen scaffold splits had zero train-test scaffold overlap; their remaining similarity and prevalence diagnostics were descriptive and did not alter the primary results.

3.6.4 Lightweight ECFP4 controls

Two lightweight ECFP4 controls were also run on the same frozen splits. Distance-weighted kNN achieved scaffold ROC AUC 0.7110 and AUPRC 0.8467, while logistic regression achieved ROC AUC 0.7063 and AUPRC 0.8542; both ROC-AUC values were below ECFP4-RF, showing that the RF result should not be generalized to every ECFP4 learner. Under the random split, the corresponding kNN and logistic ROC AUC/AUPRC values were 0.9166/0.9562 and 0.8790/0.9456, respectively. These AUPRC values apply only to the lightweight controls and extended GNN reruns; they are not retroactively assigned to the historical canonical estimates.

3.6.5 ChemBERTa extension

The extended ChemBERTa rerun is complete as a separately versioned secondary analysis with five audited partitions and 125 fold-seed records; it does not replace the prespecified benchmark.

A separate RF-only ChEMBL threshold audit was completed for four prespecified threshold specifications, with scaffold AUC ranging from 0.9271 to 0.9608; this is threshold sensitivity for the RF transfer analysis, not a full GIN threshold rerun. Paired model-minus-reference contrasts and their 95% intervals are archived in Supporting Information (Section 4). The LISH analysis is also intentionally limited: it is phenotype-only, uses aggregated MoA-associated labels, and has no drug-SMILES mapping. It therefore serves as an orthogonal pharmacological reference and control-sensitivity analysis, not as a molecular-model validation or evidence of causal mechanism (Supporting Information (Section 1)).

3.7 Limitations

The study is limited to one curated antimalarial natural-product panel, the evaluated model configurations and training budgets, 2D molecular graphs, and a scaffold split that serves only as a proxy for prospective chemical novelty. The external ChEMBL analysis is thresholded and observational rather than prospective experimental validation; it is also a transfer comparison rather than a strict replication because its split files and activity curation were constructed independently. Finally, the LISH analysis lacks a versioned drug-SMILES mapping and therefore cannot validate transfer of the molecular representations (Supporting Information (Section 1)).

3.8 Future directions

Several extensions would test questions beyond the scope of this benchmark. Post-hoc expected and maximum calibration errors and Brier scores computed on the archived test predictions are reported in the Supporting Information (Section 2); they show that calibration degrades under scaffold shift (ECE roughly doubles from the random to the scaffold split for every model family). Calibration slope/intercept fits and recalibrated-model variants would still require a new locked protocol and are not part of the present results. A pre-specified distance-to-nearest-training-compound analysis or cluster-based split could provide a stricter chemical extrapolation test; these analyses would require new locked protocols and, for cluster-based evaluation, complete retraining. Larger architectures, contrastive pre-training, latent-variable models, and conditional generation could address representation learning or candidate design, but would constitute separate experiments rather than post hoc improvements to the current comparison. Ultimately, prospective biochemical or cellular testing is required to determine whether computational rankings predict activity. None of these extensions establishes causal target engagement without independent biological evidence.

4 Conclusions

In this scaffold-controlled benchmark of 19 836 African antimalarial natural products, ECFP4-RF outperformed the tested GIN, topological-fusion GNNs, and fine-tuned ChemBERTa transformer under both random and Bemis-Murcko scaffold cross-validation. Under scaffold separation, the hierarchy was ECFP4-RF (0.8300) > GIN-TFP (0.8138) > GIN-TNE (0.8090) > GIN (0.8047) > ChemBERTa (0.7867); all four learned arms were significantly below the fingerprint baseline after multiplicity correction. The external ChEMBL-derived analysis showed the same ordering, while remaining an observational transfer analysis of thresholded activity records. Persistent-image dimensions had the largest projection-weight salience in the fusion head, indicating representation-level geometric structure without demonstrating feature necessity or improved ranking. Across the original aggregate salience summaries, the top 10

For panels comparable to the evaluated antimalarial natural-product collection, ECFP4-RF is an appropriate reference model: it is inexpensive and retained the highest ranking performance in these experiments. This recommendation is conditional on the panel, label construction, split protocol, and implementation tested here. Claims of neural improvement should be tested under

scaffold-held-out evaluation with paired uncertainty estimates. Topological and tensor-network descriptors may provide complementary descriptive information, but their inclusion should not be interpreted as a predictive gain without a direct paired comparison. Finally, pretrained transformer weights should be reset independently for every fold to prevent cross-fold information transfer. The separate LISH benchmark provides phenotype-only pharmacological context, but its lack of molecular identifiers prevents a direct structure-to-MoA comparison; its metrics are not numerically comparable to the molecular ROC-AUC results (Supporting Information (Section 1)). Experimental validation on prospectively synthesized candidates remains necessary to test whether the computational ranking predicts biological activity.

5 List of Abbreviations

ECFP4: extended-connectivity fingerprint, radius 2; **RF**: random forest; **GNN**: graph neural network; **GIN**: graph isomorphism network; **TFP**: topological fingerprint; **TNE**: tensor-network embedding; **MoA**: mechanism of action; **LISH**: public MoA benchmark; **ROC AUC**: receiver-operating-characteristic area under the curve; **CV**: cross-validation; **NP**: natural product; **ADMET**: absorption, distribution, metabolism, excretion, toxicity.

6 Methods

6.1 Data and panel

We use a frozen panel of 19 836 molecules with binary antimalarial activity labels from the eos80ch screening collection, curated and deduplicated by canonical SMILES as described previously [6]. Features are ECFP4 (2 048-bit Morgan radius-2), a 78-dimensional topological fingerprint (TFP; persistent homology, persistent images, and Betti descriptors), and 192-dimensional tensor-network embeddings (TNE). All features are aligned to the same molecule index.

Feature assembly fails rather than imputes. A SMILES string that RDKit cannot parse raises during graph construction; a non-finite value anywhere in the TNE matrix aborts the panel export; and a molecule absent from the precomputed TFP or TNE source table raises rather than receiving that column’s mean. The third guard is the one that changes results. Mean imputation would leave the affected molecules carrying panel-average topology regardless of their actual structure while still contributing to the test metric, which inflates apparent performance for exactly the molecules whose descriptors are unknown. The panel is therefore defined to admit only molecules that carry their own measured features, and the released panel contains no imputed feature value. The external ChEMBL transfer panel is the single place this policy is relaxed: there an unparseable SMILES yields an empty graph so the benchmark completes over a public download we did not curate. No canonical result depends on that path.

6.2 Splits

Two protocols were evaluated, each with five folds repeated over five seeds (25 fold–seed replicates). (i) *Random*: stratified five-fold cross-validation. (ii) *Scaffold*: molecules were grouped by Bemis–Murcko scaffold before fold assignment. A Bemis–Murcko scaffold is what remains of a molecule once every side-chain substituent is deleted: its ring systems together with the linker atoms that join them [27]. Two molecules fall in the same group when they share that framework, however differently they are substituted. Scaffolds were computed with RDKit (`MurckoScaffold.MurckoScaffoldSmiles`), and whole scaffold groups were assigned to folds by greedy minimum-maximum size balancing, so the held-out test scaffolds are disjoint from the training scaffolds. Validation molecules were sampled from the non-test portion and therefore can share scaffolds with training molecules; early stopping consequently assesses optimization

on the training chemical distribution, whereas the held-out test fold provides the scaffold-shift estimate. The split indices are released with the study.

6.3 Models

ECFP4-RF: the reference baseline, a scikit-learn `RandomForestClassifier` on the ECFP4 fingerprint matrix with 500 trees and `random_state=0`, fitted after `StandardScaler` inside the same pipeline, with scikit-learn defaults for every remaining argument including `class_weight=None`. **GIN**: graph isomorphism network (128 hidden units, 3 layers, ReLU activations, batch normalization after each layer, dropout $p = 0.1$ during training, mean-pooling readout, single head) trained on molecular graphs (atom/edge features from RDKit). We select GIN as the representative GNN architecture following MolGraphBench [11], which identified GIN as the optimal GNN for molecular regression across multiple benchmarks; GCN, GAT, and GraphSAGE were evaluated in that study and did not consistently outperform GIN, so we omit them to avoid redundancy. **GIN-TFP** / **GIN-TNE**: the same GIN with the topological descriptor vector concatenated at the head ($\text{head}_0 = \text{Linear}(128 + n_{desc} \rightarrow 128)$), enabling the salience attribution. **ChemBERTa**: pretrained SMILES transformer ('seyonec/ChemBERTa-zinc-base-v1'), fine-tuned with cross-entropy and independent pretrained-weight initialization for every fold. The GIN-family models used a maximum of 50 epochs with patience 10, batch size 512, learning rate 1×10^{-3} , and weight decay 1×10^{-4} . ChemBERTa used maximum sequence length 128, batch size 32, a maximum of 10 epochs, patience 3, learning rate 2×10^{-5} , and weight decay 0.01. All deep models ran on a single NVIDIA RTX A4000 GPU with AdamW and validation-AUC model selection.

6.4 Fold-independent transformer initialization

For each fold, the pretrained ChemBERTa state is restored before fine-tuning, so no fold receives weights updated on another fold. Without this reset, a fold’s test molecules would already have been seen during an earlier fold’s fine-tuning, and the reported test AUC would measure memorization as well as generalization. All transformer results use independent fold initialization and the same training schedule.

6.5 Orthogonal LISH MoA benchmark

The full LISH protocol, data preparation, complete metric set, and control-sensitivity table are provided in the Supporting Information (Section 1). In brief, we used the public LISH MoA training release as a phenotype-only benchmark: after drug-level aggregation it comprised 3 289 drugs and 206 scored MoA labels; because no versioned drug-SMILES mapping was available, the benchmark was not used to train or evaluate molecular fingerprints, GNNs, or transformers. An unweighted per-label logistic model was fitted independently for each label, evaluated with five random seeds and five drug-grouped folds. All fold records were audited for missing or non-finite metrics before interpretation, and the endpoint is described as MoA-associated prediction rather than causal target engagement [19].

6.6 Statistics

Per model and split, we retain the 25 fold-seed AUCs and summarize them as five per-seed means, each averaging the five fold-level test AUCs. Model-versus-baseline comparisons use paired t -tests on these five per-seed means ($df = 4$), rather than treating the 25 fold-seed values as independent observations. Table values are mean $\pm$ SD across the five per-seed means; the raw two-sided p -values are corrected with Benjamini-Hochberg separately across the four scaffold comparisons and the four random comparisons. As a distribution-free corroboration of these parametric comparisons, paired permutation tests were computed on the same five per-seed

means (10 000 sign-flip permutations under a fixed seed); they are reported alongside the t -tests in Results. Under scaffold this yields adjusted p -values of 0.033 0 (GIN), 0.033 0 (GIN-TFP), 0.037 8 (GIN-TNE), and 0.000 2 (ChemBERTa); under random all four arms remain below the baseline after correction within their own separate family. The significance threshold is $\alpha = 0.05$ and effect sizes are reported as mean differences. Saliency is aggregated across the 25 replicates as the per-dimension mean absolute projection weight.

Availability of data and materials

All code, frozen splits (random and scaffold, five seeds each), per-fold results, saliency data, figure scripts, and the 19 836-molecule panel are available under the MIT license in the public GitHub repository (https://github.com/NanaEngo/Malaria_codesV2). The audited LISH phenotype-only fold outputs, comparison report, and the full Supporting Information (Section 1) are also released in the repository’s P5 results artefacts. The official challenge source, derived-data provenance, and archive hash are recorded in the accompanying LISH README; the raw public archive and any credentials are not redistributed. Trained model states and feature tables are included; reproducibility is supported by the frozen split files and the pinned environment. **Secondary scaffold-only replication.** A repeat scaffold evaluation of the ChemBERTa arm (five folds $\times$ five seeds, same code and same frozen splits as the primary run, distinct training run) produced all 25 fold-seed records and a mean ROC AUC of $0.790\,8 \pm 0.005\,8$ when summarized across five per-seed means; the primary scaffold estimate was $0.786\,7 \pm 0.005\,4$ under the same summary. The primary random-split estimate was $0.912\,1 \pm 0.001\,2$; no repeat random-split run was performed.

Author contributions

MVST conceived the study, designed the benchmark protocol, implemented the GNN/transformer pipelines, and wrote the manuscript. All authors contributed to data analysis, reviewed the results, and approved the final version. (Full author affiliations are given on the title page.)

Competing interests

The authors declare no competing interests.

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Ethics approval and consent to participate

Not applicable. This study is entirely computational and uses publicly available chemical databases and in-silico activity predictions; no human or animal subjects or patient data were involved.

Consent for publication

Not applicable. No individual person’s data (including images, videos, or identifiable details) appear in this manuscript.

Use of Artificial Intelligence

The authors used AI-assisted tools during code development and data-analysis scripting. All affected text and code were reviewed and edited by the authors, who take full responsibility for the article. No AI system contributed as an author. The reported computations, statistical analyses, and quantitative claims were generated and verified by the authors.

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